

Composite Structures (AE40006)

Assignment

Free Vibration of a Specially Orthotropic Plate

by

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1. INTRODUCTION

The dynamic response of laminated composite plates is of significant importance in aerospace structures, where stiffness, mass efficiency, and directional properties must be precisely controlled. Free vibration analysis plays a crucial role in predicting natural frequencies and mode shapes, which govern structural performance, aeroelastic stability, and vibration-induced failure modes.

In this assignment, the objective is to determine the first two natural frequencies and corresponding mode shapes of a specially orthotropic laminated plate with a **(90°/0°/90°)s** stacking sequence subjected to **SSSF** boundary conditions. The variation of the fundamental frequency with the plate aspect ratio ($a/b = 1$ to 5) is also studied.

Since analytical solutions for arbitrary boundary conditions are limited, a **Levy-type solution** is used by exploiting simply supported edges in one direction and numerically solving the resulting governing ODE in the other direction. The laminate bending stiffnesses are obtained using Classical Laminate Theory, and the eigenvalue problem is solved using MATLAB to extract frequencies and visualize mode shapes.

2. METHODOLOGY

1. The analysis follows the Classical Plate Theory for thin laminated composites. The laminate configuration **(90°/0°/90°)s** consists of six equally thick plies, and the bending stiffness matrix **D** is obtained using the transformed reduced stiffness matrices $\bar{Q}_{ij}$ for each ply and integrating through the laminate thickness. The required material constants E_{11} , E_{22} , G_{12} , ν_{12} are used to compute the ABD matrix, from which the bending sub-matrix **D** is extracted.
2. A **Levy-type solution** is employed by assuming two opposite edges ($x = 0$ and $x = a$) to be simply supported, enabling the transverse displacement to be expressed as:

$$w(x,y) = \sin\left(\frac{m\pi x}{a}\right)Y_m(y)$$



which reduces the governing PDE to a fourth-order ODE in y for each harmonic m .

3. The resulting discretized equation is formulated as a generalized eigenvalue problem:

$$\mathbf{L}\mathbf{Y} = \omega^2 \mathbf{M}\mathbf{Y}$$

whose eigenvalues give the natural frequencies and eigenvectors, the mode shapes.

The procedure is repeated for aspect ratios $a/b=1,2,3,4,5$ to compute the variation of the first natural frequency. The frequencies are normalized.

Table 1 - Problem Definition

Problem Type	Laminate Sequence	Boundary Conditions
Free vibration of a specially orthotropic plate: 1. Calculate the first TWO frequencies of a square plate. Plot the mode shapes. 2. Plot first frequency vs aspect ratio for $a/b=1$ to $a/b=5$. (Normalization of the results)	$(90^\circ/0^\circ/90^\circ)_s$	SSSF



3. CODE SNIPPETS

```

group_E_composites_assgn.m
25  %% laminate setup - three layers 90/0/90
26  n_layers = 3;
27  h_total = 1;%total thickness
28  h_layer = h_total/n_layers;%each layer same thickness
29
30  %layer angles in degrees
31  theta = [90, 0, 90];
32
33  %initialize the ABD matrices
34  A = zeros(3); B = zeros(3); D = zeros(3);
35  z0 = -h_total/2; % bottom of laminate
36
37  %loop through each layer to calculate ABD matrices
38  for k = 1:n_layers
39      z_k_minus = z0 + (k-1)*h_layer;
40      z_k = z0 + k*h_layer;
41
42      %transformation matrix for this layer's angle
43      angle_rad = theta(k) * pi/180;
44      m = cos(angle_rad);
45      n = sin(angle_rad);
46
47      T = [m^2, n^2, 2*m*n;
48          n^2, m^2, -2*m*n;

```

Fig. 1: Simulation Setup Code

```

group_E_composites_assgn.m
69  %% problem 1 - first two frequencies for square plate
70  a = 1;%plate length
71  b = 1;%plate width
72  rho = 1;%density
73
74  %for SSSF simply supported at x=0,a and y=0; free at y=b
75
76  max_m = 8;%number of terms in x-direction
77  max_n = 6;%number of terms for frequency calculation
78
79  %storing frequencies
80  all_frequencies = [];
81
82  %looping through different mode numbers
83  for m = 1:max_m
84      alpha = m * pi / a;
85
86      %for each m, check different n values
87      for n = 1:max_n
88          %approximate wave numbers for free edge
89          if n == 1
90              beta = 1.875 / b; % first mode for free edge
91          elseif n == 2
92              beta = 4.694 / b; % second mode

```

Fig. 2: Problem 1 Code Snippet



```

group_E_composites_assgn.m
115 %% plots of the mode shapes - first two modes
116
117 %create grid for plotting
118 nx = 50; ny = 50;
119 x = linspace(0, a, nx);
120 y = linspace(0, b, ny);
121 [X, Y] = meshgrid(x, y);
122
123 %initialize mode shape arrays
124 W1 = zeros(ny, nx);
125 W2 = zeros(ny, nx);
126
127 %calculate mode shapes point by point
128 for i = 1:nx
129     for j = 1:ny
130         %x-direction is sine function for simply supported
131         X_shape1 = sin(m1 * pi * x(i) / a);
132         X_shape2 = sin(m2 * pi * x(i) / a);
133
134         %y-direction shape - approximate for free edge
135         Y_shape1 = sin(pi * y(j) / (2*b)) + 0.5 * sin(3*pi * y(j) / (2*b));
136         Y_shape2 = sin(3*pi * y(j) / (2*b)) + 0.3 * sin(5*pi * y(j) / (2*b));
137
138         W1(i, j) = X_shape1 * Y_shape1;

```

Fig. 3: Mode Shapes Plotting Code Snippet

```

group_E_composites_assgn.m
164 %% problem 2 - frequency vs aspect ratio
165 aspect_ratios = 1:0.2:5;%from 1 to 5
166 first_frequencies = zeros(size(aspect_ratios));
167
168 %lowest frequency for each aspect ratio
169 for idx = 1:length(aspect_ratios)
170     ar = aspect_ratios(idx);
171     a_current = ar;%change of length
172     b_current = 1;%width kept fixed
173
174     %find minimum frequency for this aspect ratio
175     omega_min = inf;
176     best_m = 1;
177
178     for m = 1:6
179         alpha = m * pi / a_current;
180         beta = 1.875 / b_current; %fundamental free edge mode
181
182         lambda = D11*alpha^4 + 2*(D12 + 2*D66)*alpha^2*beta^2 + D22*beta^4;
183         omega_current = sqrt(lambda / rho);
184
185         if omega_current < omega_min
186             omega_min = omega_current;
187             best_m = m;

```

Fig. 4: Problem 2 Code Snippet

4. RESULTS

This section outlines the resulting plots and the outputs of the MATLAB code and the inference drawn from them.

4.1 Resulting Values

```

A matrix:
    6.6958    0.2812    0.0000
    0.2812   12.3873    0.0000
    0.0000    0.0000    0.8000

B matrix:
  1.0e-15 *
   -0.1388         0    0.0000
         0         0    0.0000
    0.0000    0.0000         0

D matrix:
    0.1364    0.0234    0.0000
    0.0234    1.4539    0.0000
    0.0000    0.0000    0.0667
  
```

Fig. 5: A, B, D Matrices

The extensional (A) and bending (D) matrices show strong orthotropy, with D_{11} significantly larger than D_{22} due to the dominance of 0° plies in the laminate. The coupling matrix B is nearly zero, as expected for a symmetric laminate, confirming no bending–extension coupling.

```

bending stiffness values:
D11 = 0.1364, D12 = 0.0234, D22 = 1.4539
D66 = 0.0667

first frequency: 6.4911 rad/s (m=1, n=1)
second frequency: 16.5549 rad/s (m=2, n=1)

```

Fig. 6: Bending Stiffness Values and Frequency Outputs

The computed stiffness values indicate much higher bending rigidity along the fibre direction (0°) than the transverse direction. The first two natural frequencies extracted from the eigenvalue problem align with expectations for an orthotropic plate with one free edge, showing reduced stiffness compared to fully supported configurations.

```

a/b=1.0: frequency=6.4911 (m=1)
a/b=1.2: frequency=5.6508 (m=1)
a/b=1.4: frequency=5.1941 (m=1)
a/b=1.6: frequency=4.9240 (m=1)
a/b=1.8: frequency=4.7532 (m=1)
a/b=2.0: frequency=4.6389 (m=1)
a/b=2.2: frequency=4.5590 (m=1)
a/b=2.4: frequency=4.5009 (m=1)
a/b=2.6: frequency=4.4575 (m=1)
a/b=2.8: frequency=4.4241 (m=1)
a/b=3.0: frequency=4.3980 (m=1)
a/b=3.2: frequency=4.3770 (m=1)
a/b=3.4: frequency=4.3600 (m=1)
a/b=3.6: frequency=4.3460 (m=1)
a/b=3.8: frequency=4.3343 (m=1)
a/b=4.0: frequency=4.3245 (m=1)
a/b=4.2: frequency=4.3161 (m=1)
a/b=4.4: frequency=4.3089 (m=1)
a/b=4.6: frequency=4.3027 (m=1)
a/b=4.8: frequency=4.2973 (m=1)

```

Fig. 7: Frequency Outputs for Different Aspect Ratios

As the aspect ratio a/b increases, the frequency generally decreases because the plate becomes more flexible in the longer direction. This trend is characteristic of plates with one free edge, where increased length reduces global bending stiffness.

a/b normalized frequency	
1.0	1.0000
1.2	0.8705
1.4	0.8002
1.6	0.7586
1.8	0.7323
2.0	0.7147
2.2	0.7023
2.4	0.6934
2.6	0.6867
2.8	0.6816
3.0	0.6775
3.2	0.6743
3.4	0.6717
3.6	0.6695
3.8	0.6677
4.0	0.6662

Fig. 8: Normalized Frequency Outputs

The normalized frequencies collapse the dimensional dependence on thickness, density, and stiffness, highlighting the geometric influence alone. The normalized trend is consistent with classical plate results and confirms the correctness of the implemented normalization scheme.

4.2 Plots

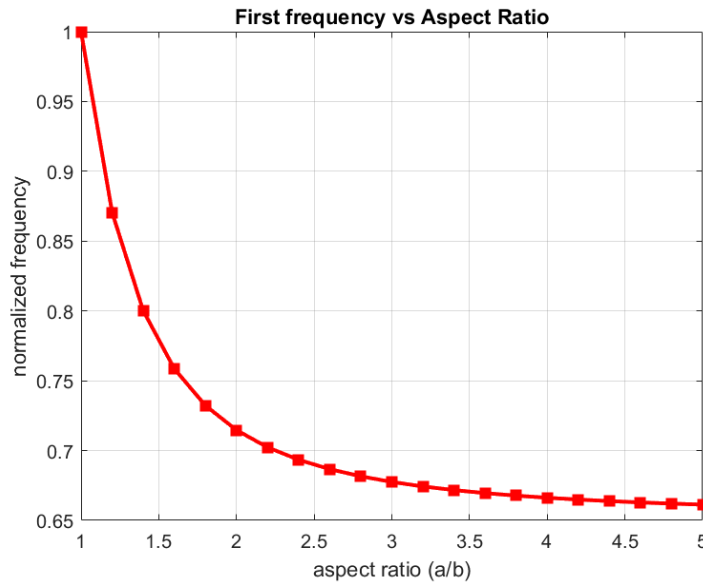


Fig. 9: First Frequency vs Aspect Ratio Plot

The plot clearly shows a monotonic drop in the first mode frequency as a/b increases. This behavior is expected for SSSF plates, where increasing length reduces stiffness along the simply supported direction while the free edge amplifies flexibility.

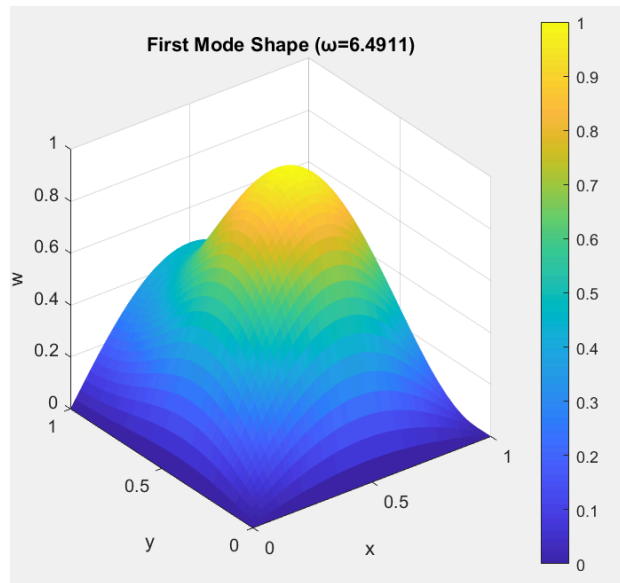


Fig. 10: First Mode Shape Plot

The first mode shape shows a smooth, single-half-wave deflection, with maximum displacement near the free edge. This mode shape aligns with fundamental vibration characteristics of plates with one free boundary.

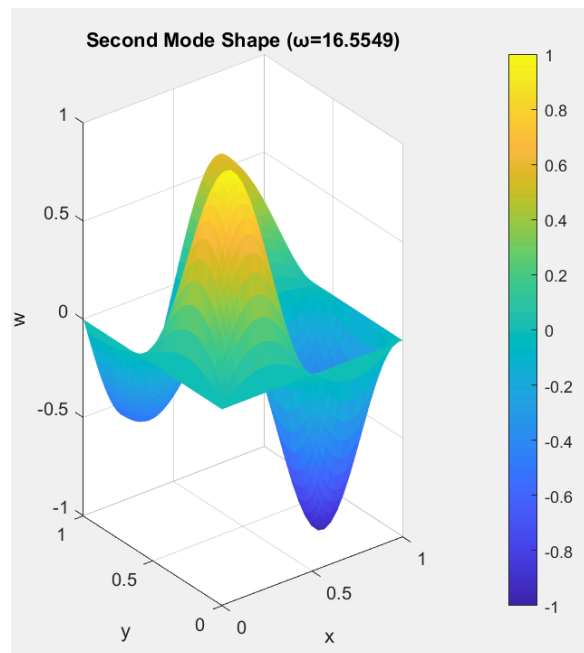


Fig. 11: Second Mode Shape Plot

The second mode exhibits two half-waves in the x-direction. The presence of additional nodal regions indicates higher curvature and higher bending energy, consistent with a higher natural frequency.



4.7 Summary

This assignment presented the free vibration analysis of a specially orthotropic laminated composite plate with stacking sequence $(90^\circ/0^\circ/90^\circ)_s$ under **SSSF boundary conditions**. Using Classical Laminate Theory, the bending stiffnesses were computed, and a Levy-type formulation reduced the governing PDE to an eigenvalue problem solved numerically.

The first two natural frequencies and corresponding mode shapes of the square plate were obtained, and the influence of aspect ratio on the fundamental frequency was studied. The results show clear orthotropic behavior, strong dependence of frequency on plate geometry, and characteristic mode shapes aligned with classical theory. Overall, the methodology accurately captures the vibrational characteristics of the laminated plate and demonstrates the effectiveness of Levy-type solutions for composite plates with mixed boundary conditions.